

Retail Customer Intelligence & Churn Prediction Report

This report presents an end-to-end analysis of retail transaction data, enhanced with machine learning-based churn prediction to enable data-driven decision making.

Business Problem

The business lacks visibility into customer retention, revenue drivers, and behavioral patterns, limiting its ability to make proactive decisions and improve long-term growth.

Objectives

- Analyze revenue trends and seasonality
- Identify top-performing products and customers
- Understand customer behavior and segmentation
- Predict customer churn using machine learning
- Provide actionable recommendations

Methodology

- Data cleaning and preprocessing using Python
- SQL-based analysis for business insights
- Feature engineering using RFM (Recency, Frequency, Monetary)
- Churn prediction using Logistic Regression and Random Forest
- Dashboard development using Power BI

Key Metrics

Total Revenue: 8.6M+
Total Orders: 19K+
Total Customers: 4.3K+
Average Order Value: 167+

Key Insights

- Revenue shows strong seasonal trends
- A small number of products drive majority of sales
- Most customers are one-time buyers indicating high churn risk
- High-value customers contribute disproportionately to revenue

Churn Model Insights

- Customers with high recency and low frequency are most likely to churn
- Early identification enables targeted retention strategies
- Predictive insights improve marketing efficiency

Business Recommendations

- Implement customer retention and loyalty programs
- Target high-risk customers with personalized offers
- Focus on high-performing products
- Leverage seasonal trends for campaigns

Conclusion

Combining analytics with machine learning enables the business to move from reactive reporting to proactive decision-making, improving retention and revenue growth.